

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Advanced Data Structure [DS_5025]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Derive the **worst-case time complexity** of the Manipal function (5)

A)

```
void Manipal(int n) {  
    int i = 1;  
    while (i < n) {  
        for (int j = 0; j < i; j++) {  
            printf("%d ", j);  
        }  
        i *= 2;  
    }  
}
```

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Use the recursive tree method to solve the recurrence $T(n) = 2T(n/2) + cn$ and find its time complexity. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) Given the following array of numbers (5)
38, 27, 43, 3, 9, 82, 10

A) Sort the numbers using a divide-and-conquer-based Sorting algorithm with $O(n \log n)$ worst-case time complexity. Write the Pseudocode for the algorithm.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) You are given the following sequence of integers to insert into an initially empty AVL Tree (5)
50, 30, 70, 20, 40, 60, 80, 10, 35, 45

- Insert the nodes one by one into the AVL Tree.
- After each insertion, perform necessary rotations to maintain AVL balance.
- Draw the final AVL Tree structure.
- Mention the balance factor of each node in the final tree.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3) Given a chain of matrix (5)
A1: 5X4
A2: 4X3

- A) A4: 5X2
A3: 3X5
A5: 2X4

where A_i has dimensions $p_{i-1} \times p_i$. Fully parenthesize the product $A_1 . A_2 . \dots . A_5$ in a way that minimizes the number of scalar multiplications.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Given an array consisting of the following elements: {-2, -5, 6, -2, -3, 1, 5, -6}. Find a contiguous non-empty subarray within the array that has the maximum sum, and return the sum as well as the subarray elements. Implement using an appropriate design paradigm, give the optimized algorithm, and derive the time complexity. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4) Differentiate between FIFO and LIFO batch and bound methods for the following 0/1 knapsack problem, where the knapsack capacity is 15kg. (5)

A)

Item	X1	X2	X3	X4	X5
Profit	5	7	9	10	12
Weight (kg)	2	5	7	8	10

Illustrate the FIFO and LIFO-based knapsack solution trees.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Use a Multiobjective Genetic Algorithm to solve a problem with two conflicting objectives. Apply Pareto optimality to select the best solutions from the population. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Use a polynomial-time reduction to show that a given problem is NP-complete. Apply the two NP-completeness conditions and demonstrate the process using a known NP-complete problem as the basis for reduction. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Perform a polynomial-time reduction from the 3-SAT problem to the Clique problem. Apply the transformation to a sample 3-SAT instance and construct the corresponding graph to demonstrate how satisfiability is preserved." (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_5013:Advanced Data Structure

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_5013:Advanced Data
Structure

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)